THEORY OF COMPUTATION

EXPERIMENTS 21 - 25

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21. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0A1 \qquad A \rightarrow 0A \mid 1A \mid \varepsilon$

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, n, flag = 1;

    scanf("%99s", s);
    n = strlen(s);

    if (n < 2 || s[0] != '0' || s[n-1] != '1')
        flag = 0;

    for (i = 0; i < n; i++) {
        if (s[i] != '0' && s[i] != '1')
            flag = 0;
    }

    if (flag)
        printf("Accepted");
    else
        printf("Rejected");

    return 0;
}
```

22. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \varepsilon$

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, j, flag = 1;

    scanf("%99s", s);

    i = 0;
    j = strlen(s) - 1;

    while (i < j) {
        if (s[i] != s[j]) {
            flag = 0;
            break;
        }
        i++;
        j--;
    }

    if (flag)
        printf("Accepted");
    else
        printf("Rejected");

    return 0;
}
```

23. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S0 \mid A \quad A \rightarrow 1A \mid \varepsilon$

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i = 0, j, flag = 1;

    scanf("%99s", s);
    j = strlen(s) - 1;

    while (i < j && s[i] == '0' && s[j] == '0') {
        i++;
        j--;
    }

    for (; i <= j; i++) {
        if (s[i] != '1') {
            flag = 0;
            break;
        }
    }

    if (flag)
        printf("Accepted");
    else
        printf("Rejected");

    return 0;
}
```

24. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S1 \mid \varepsilon$

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, n, count0 = 0, count1 = 0, flag = 1;

    scanf("%99s", s);
    n = strlen(s);

    for (i = 0; i < n; i++) {
        if (s[i] == '0' && count1 == 0)
            count0++;
        else if (s[i] == '1')
            count1++;
        else
            flag = 0;
    }

    if (count0 != count1)
        flag = 0;

    if (flag)
        printf("Accepted");
    else
        printf("Rejected");

    return 0;
}
```

25. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow A101A, \quad A \rightarrow 0A \mid 1A \mid \varepsilon$

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, flag = 0;

    scanf("%99s", s);

    for (i = 0; s[i] != '\0'; i++) {
        if (s[i] == '1' && s[i+1] == '0' && s[i+2] == '1') {
            flag = 1;
            break;
        }
    }

    if (flag)
        printf("Accepted");
    else
        printf("Rejected");

    return 0;
}
```